

Aroosh Datta

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LinkedIn — GitHub — LeetCode

EDUCATION

Vellore Institute of Technology

Bachelor of Engineering in Computer Science; GPA: 8.33

Bhopal, Madhya Pradesh

Sep 2024 - Present

Lucknow Public School

Intermediate (Class XII); Percentage: 89.57%

Lucknow, India

2023 - 2024

COMPUTATIONAL PROBLEM SOLVING

LeetCode | Rating: 1503 (Max)

Consistent Solver

- Solved **230+** Data Structures & Algorithms problems, focusing on Graphs and Dynamic Programming.
- Maintained proficiency in Python for algorithmic efficiency and memory optimization.

ACHIEVEMENTS

- 3rd Position**, Rapid Coding Contest (Mozilla Firefox Club): Optimized algorithms under 45 mins.
- Top Performer**, Dawn of Code (MERN Stack Club): Solved complex backend integration challenges.
- Participant**, Summer of Code Fest '25 (GSoC Club): Contributed to open-source problem statements.

TECHNICAL SKILLS

Languages: Python (Advanced), C++, Java, SQL, JavaScript

Core Concepts: Data Structures & Algorithms (DSA), System Design, OOP, Distributed Systems

Frameworks/Libraries: Flask, React, TensorFlow, Redis, Tkinter

Tools: Git/GitHub, Linux (Bash), Docker, MySQL

PROJECTS

Distributed Task Queue | Python, Redis, Multiprocessing

- Engineered a fault-tolerant **asynchronous task execution system** capable of distributing workloads across multiple worker nodes, similar to Celery.
- Implemented **task persistence** and retry logic using Redis to ensure 99.9% reliability during node failures.
- Optimized job scheduling algorithms, reducing task processing latency by 40% compared to serial execution.

PySticky | Python, Tkinter, SQLite

- Developed a lightweight, persistent sticky-note application with **local storage capabilities**, allowing users to organize tasks efficiently.
- Engineered a custom drag-and-drop interface and implemented auto-save functionality to prevent data loss.
- Designed a modular codebase following **OOP principles**, ensuring scalability for future feature additions.

Skin Lesion Diagnosis | Python, TensorFlow, CNN

- Collaborated in a team of 5 to build a **Computer Vision model** for early detection of melanoma and skin cancer.
- Trained a **Convolutional Neural Network (CNN)** on the HAM10000 dataset, achieving high validation accuracy.
- Preprocessed over 10,000 dermoscopic images using OpenCV to reduce noise and improve model generalization.

POSITIONS OF RESPONSIBILITY

BashCraft Club (Linux & DevOps)

Technical Content Strategist

VIT Bhopal

Dec 2024 - Present

- Curated technical blogs and documentation on Linux commands and shell scripting, simplifying complex concepts for **500+ student members**.
- Spearheaded the content strategy for the club's "Bash Series," increasing social media engagement by 20%.

Fine Arts Club

Creative Lead

VIT Bhopal

Feb 2025 - Present

- Managed event documentation and digital content for major college cultural fests, coordinating with a team of 10+ members.
- Streamlined communication channels between the design and executive teams, ensuring timely project delivery.